



**AnalystLab**

**Africa - Data**

**Science Internship Programme**

**Week 3 Assignment**

## **Business Insights Report**

**Advanced Data Exploration & Feature Engineering  
Loan Prediction Dataset**

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**Internship:** Data Science

**Batch:** D

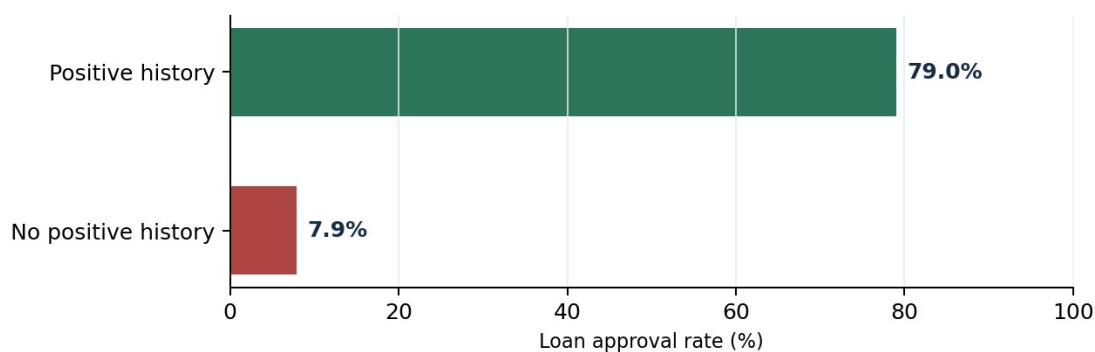
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## Executive summary

This report evaluates 614 historical loan applications to identify the factors most closely associated with approval and to prepare a reliable foundation for predictive modelling. Of these applications, 422 were approved and 192 were rejected, corresponding to an approval rate of 68.73%.

**Principal finding** Applicants with positive credit history recorded a 79.05% approval rate, compared with 7.87% for applicants without positive credit history.

| Portfolio measure            | Result    |
|------------------------------|-----------|
| Applications analysed        | 614       |
| Approved / rejected          | 422 / 192 |
| Overall approval rate        | 68.73%    |
| Training / test observations | 491 / 123 |



## Business context and analytical approach

Lending institutions must balance access to credit with consistent risk assessment. The project examined whether applicant characteristics, household income, requested loan size and property area show meaningful relationships with historical approval decisions. The analysis combined data-quality validation, exploratory visualization, hypothesis testing, feature engineering and leakage-safe preparation for machine learning.

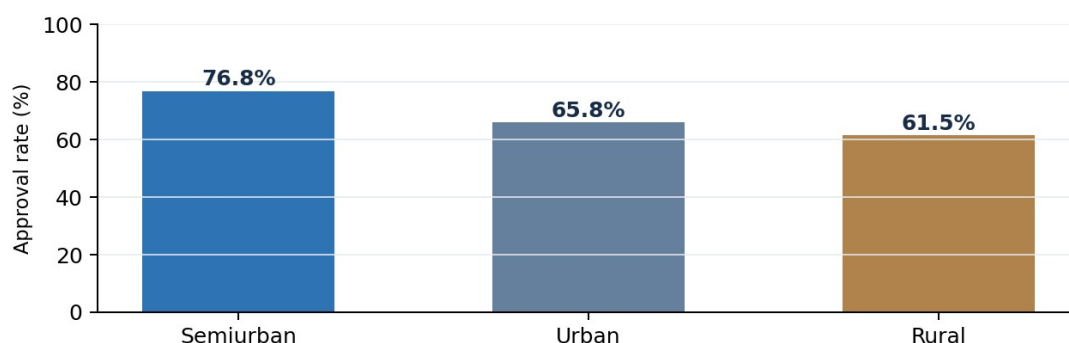
## Key findings

### Credit history is the dominant observed factor

The relationship between credit history and approval was statistically significant (chi-square = 176.11;  $p < 0.001$ ). Cramer's V of 0.536 indicates a materially stronger association than the other tested categorical variables. Credit information should therefore remain central to subsequent modelling and data-quality controls.

### Property area provides a weaker secondary signal

Approval rates were highest in semiurban areas (76.82%), followed by urban areas (65.84%) and rural areas (61.45%). The association was statistically significant ( $p = 0.002$ ), but its effect size was modest (Cramer's V = 0.142). Geographic differences warrant investigation before they influence operational policy.



## What the financial variables reveal

Total household income and requested loan amount did not differ significantly between approved and rejected applications when assessed separately. Total income yielded  $p = 0.713$ , while loan amount yielded  $p = 0.398$ . These results do not mean financial information is irrelevant; rather, raw values alone do not explain historical approval outcomes in this dataset.

Total income and requested loan amount were strongly related (Spearman  $\rho = 0.688$ ;  $p < 0.001$ ). Higher-income households generally requested larger loans. This supports analysing affordability as a relationship between income, loan amount and repayment term rather than as an isolated threshold.

## Feature-engineering decisions

- o Payment-income ratio: combines estimated monthly principal, household income and repayment duration into an affordability proxy.
- o Coapplicant-income share: measures how much household income depends on a second applicant.
- o Dependents and household size: convert family information into interpretable numerical or reporting categories.
- o Log-transformed income and loan amount: reduce the influence of extreme values and improve modelling stability.
- o Credit-risk and income bands: provide readable stakeholder categories while avoiding unnecessary duplication in final predictive inputs.

## Recommended management actions

1. Prioritize the completeness and verification of credit-history information.
2. Assess affordability through combined repayment and income indicators.
3. Investigate geographic differences for product, branch-policy or data-collection effects.
4. Monitor demographic fairness without using gender as a predictive input.

## Limitations and responsible interpretation

- o The dataset contains 614 applications; estimates for smaller applicant groups may be unstable.
- o Historical approval is the target, not repayment performance or actual default risk.
- o Interest rates, existing debt, expenses and verified monthly instalments are unavailable.
- o Previously imputed values may reduce natural variation in some variables.
- o Application dates and applicant identifiers are absent, limiting temporal and repeat-applicant analysis.
- o Observed associations are not proof of causation and may reflect historical institutional bias.

## Next steps for predictive modelling

- o Train a baseline classifier and compare logistic regression, decision tree and random-forest approaches.
- o Use stratified cross-validation and evaluate precision, recall, F1-score, ROC-AUC and confusion matrices.
- o Preserve the predetermined training/test split and keep encoding and scaling fitted on training data only.
- o Audit subgroup fairness, evaluate probability calibration and document decision-threshold trade-offs.

## Conclusion

The Week 3 investigation transformed a cleaned application dataset into a statistically examined, feature-engineered and modelling-ready resource. Credit history is the clearest observed approval factor, while property area contributes a smaller signal and financial variables become more informative when combined into affordability-oriented representations.